

Vision Desk Case

Computer Vision in Finance

Master's in Finance, Data & AI — student brief

Dhiya Lecheheb

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1 Question

Closed task. You receive a financial image (chart screenshot or document page). Build a **Vision Desk**: fine-tune a ViT, expose it through MCP tools, and let an LLM answer *only* from those tools.

Your deliverable must make three things explicit:

1. **Vision assertion** — class, score, structured fields the model can support;
2. **Tool justification** — which MCP call and JSON field justify each claim;
3. **Refusal** — what you do not conclude when a tool fails or returns null.

Parallel with Signal Desk. Where Signal Desk judged an LSTM *after trading costs*, Vision Desk judges an LLM agent *after grounding*.

2 Setup

Item	v1 choice
Input	20–40 labeled images: chart type <i>or</i> page type (group choice, fixed in class).
Vision model	Pretrained Vision Transformer (ViT), fine-tuned. No LSTM.
Vision output	Class + score, then JSON matching a schema (3–6 fields).
Agent	LLM + local MCP server (read-only tools).

3 Required MCP tools

Tool	Does	Does not
<code>vision.classify</code>	Class + score	Interpret the market
<code>vision.extract</code>	JSON fields	Invent a missing amount
<code>market.lookup</code>	Quote / ticker meta	Read the image
<code>knowledge.search</code>	Glossary / method note	Replace vision

Stubs are accepted for `market.*` and `knowledge.*` if they are documented and deterministic.

4 Mandatory protocol

1. Split by **document** (or issuer), not a random mix of pages from the same PDF.
2. Backbone preprocess as prescribed by the model (ImageNet mean/std or checkpoint docs).
3. Publish F1 (macro or per class) out of sample + confusion matrix.
4. Publish the JSON schema *before* test inference.
5. Every claim in the LLM answer must cite a tool call and a field.
6. Force a tool-failure scenario and show refuse / retry (not invention).
7. Forbidden: LLM answer without a tool for a fact “seen” in the image.

5 What you must show

- ViT vs session-1 baseline, with a clean document-level split;
- valid JSON on the held-out set (invalid schema = pipeline failure);
- one **grounded** MCP trace (tools → JSON → final answer);
- one **refusal** after tool failure or null field;
- three written limits (layout shift, OCR / capture, hallucination, ...).

6 Deliverables

Deliverable	Content
Notebook / code	Data, ViT, JSON schema, minimal MCP server, both scenarios.
Note, 4 pages	Protocol, vision metrics, grounded trace, refusal, three limits.
Oral, 8–10 min	One image, one tool trace, one limit. Defend grounding.

7 How results are read

Grading focuses on protocol, grounding, and limits. A clear refusal after tool failure scores better than a polished LLM sentence with no JSON.

8 Variant — Doc Lens

Same science. Deliverable = small club tool: image upload, MCP call, exported page (class, JSON, tools, three limits). No chat without a trace.

9 Video resources (preparation)

- StatQuest, *CNNs — image classification* — youtube.com/watch?v=HGwBXDKFk9I;
- 3Blue1Brown, *Attention in transformers* — youtube.com/watch?v=eMlx5fFNoYc;
- Yannic Kilcher, *ViT paper explained* — youtube.com/watch?v=TrdevFK_am4;
- Vision Transformer in PyTorch — youtube.com/watch?v=ovB0ddFtzzA;

- MCP (Anthropic) — youtube.com/watch?v=CQywdSdi5iA;
- MCP in Python for beginners — youtube.com/watch?v=LYfr7qusVSs.

10 Checklist

- ☐ Image set and classes fixed in class
- ☐ ViT (not LSTM) + split by document
- ☐ Baseline before the ViT
- ☐ JSON schema written before the test
- ☐ Four MCP tools (stubs OK if documented)
- ☐ Grounded trace + tool-failure refusal
- ☐ Notebook + 4-page note + oral